

HOW TO DO IT

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Work out where the problem is in the brainstem using 'the rule of 4'

Peter Gates

The brainstem is incredibly complex. It contains nuclei, pathways and structures, many with unusual names such as the olives, medial longitudinal fasciculus and the superior and inferior colliculi. The neurological examination can only assess some of them. This article is designed to help make sense of the signs in the more common brainstem syndromes using 'the rule of 4'.

INTRODUCTION

Imagine that you are examining a patient with a brainstem problem and you are trying to relate the symptoms and signs to the brainstem anatomy and blood supply. First you need a road map. The brainstem is divided into the midbrain, pons and medulla. The lower four cranial nerves (9, 10, 11 and 12) emerge from the medulla, the middle four from the pons (5, 6, 7 and 8) and the upper four cranial nerves are above the pons (the third and fourth emerge from the midbrain while the first and second are outside the brainstem) (figure 1).

The blood supply to the brainstem comes from branches arising from the vertebral and basilar arteries that supply the paramedian territory (just to the side of the midline on each side), and larger circumferential branches that supply the lateral part of the pons and medulla. This is reflected in the distinct clinical features of ischaemic events in these vascular territories.

It is the presence of vertigo or diplopia *in association with* weakness and/or sensory

disturbance in the limbs and/or face that categorically indicates a brainstem problem. *Isolated* vertigo and *isolated* diplopia are only rarely caused by intrinsic disorders of the brainstem. Although dysarthria, dysphagia, ataxia and bilateral symptoms and signs in the face and limbs occur with brainstem lesions, some of these features can also occur with lesions outside the brainstem (eg, non-dominant hemisphere problems, cerebellar disturbances and bilateral anterior cerebral artery problems) and therefore do not definitively indicate brainstem pathology.

In the examination, a cranial nerve palsy with weakness and/or sensory signs in the face or limbs, or crossed sensory signs (see below for an explanation), also strongly point to the brainstem as the site of the problem (with the notable exception of carotid dissection where unilateral cranial neuropathies and contralateral pyramidal signs may rarely be observed because the cranial nerves below the skull can be directly affected by the dissection).¹

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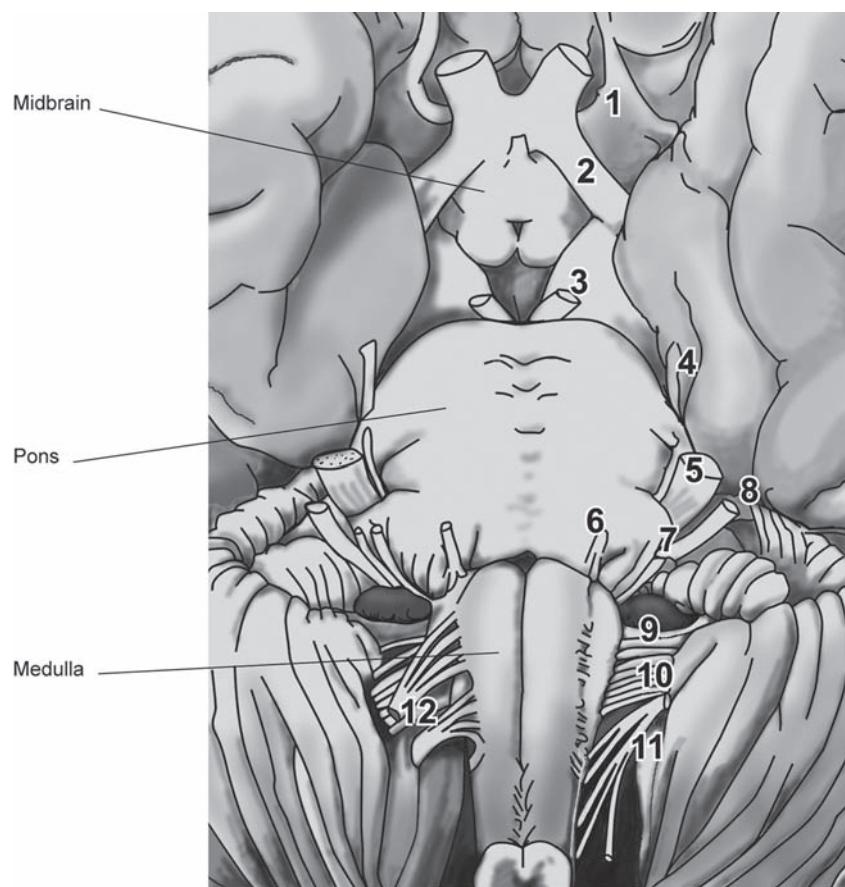


Figure 1
The ventral aspect of the brainstem demonstrating the position of the individual cranial nerves (numbered) in relation to the medulla, pons and midbrain. 1, olfactory; 2, ophthalmic; 3, oculomotor; 4, trochlear; 5, trigeminal; 6, abducent; 7, facial; 8, auditory; 9, glossopharyngeal; 10, vagus; 11, spinal accessory; 12, hypoglossal (reproduced from Gates⁴).

ANATOMY OF THE PARAMEDIAN BRAINSTEM

A simple way to remember the four structures that are in the paramedian brainstem is that they all begin with the letter 'M' (figures 2):

- Motor pathway**
- Medial lemnisci**
- Motor nuclei**
- Medial longitudinal fasciculus**

The motor pathway (usually referred to as the corticospinal or pyramidal tract) arises in the cortex and descends via the corona radiata, internal capsule and crus cerebri (cerebral peduncle) into the brainstem where it lies just to the side of the midline (paramedian) in the lower midbrain, pons and medulla. The pathway then crosses the midline in the decussation of the pyramids at the level of the foramen magnum.

The pathways for vibration and proprioception ascend in the dorsal columns of the spinal cord to the level of the foramen magnum where they cross the midline (note at the same level as the motor pathway) and then ascend up through the paramedian brainstem in the medial lemnisci to the thalamus.

The four cranial nerve motor nuclei that lie in the paramedian brainstem, dorsal to the medial

longitudinal fasciculus, are those that divide evenly into 12 (except one and two)—that is, 3, 4, 6 and 12; 3 and 4 are in the midbrain, 6 is in the pons and 12 is in the medulla.

The medial longitudinal fasciculus also lies in the paramedian brainstem but dorsal to the medial lemniscus and ventral to the motor nuclei of the third, fourth and sixth cranial nerves.

PATIENT NO 1

Let us assume the patient has a hemiparesis. You examine the motor system, followed by the sensory system, then look for cerebellar signs and then, if indicated, examine the cranial nerves. If you find upper motor neuron signs in the arm and leg on one side you know that the problem is either in the spinal cord ipsilateral to the weakness above the level of C5 or on the contralateral side of the brain above the foramen magnum (ie, above the decussation of the pyramids). If the patient also has an upper motor neuron facial weakness (the forehead is not affected) on the same side as the weakness in the limbs, the lesion may be higher in the cerebrum or, if the pathology is in the brainstem, above the seventh nerve nucleus contralateral to the weakness. In the absence of any other signs, this is as far as we can take it to localise the origin of the hemiparesis.

You then turn your attention to the sensory system to consider two pathways: one that conveys vibration and proprioception, and the other pain and temperature. If you detect abnormalities of vibration and proprioception in the limbs on the same side as the weakness, the problem could still be in the contralateral paramedian brainstem but once again you cannot definitively localise it to the brainstem because it might be ipsilateral in the high cervical spinal cord or contralateral above the midbrain. The combination of hemiparesis with or without associated altered vibration and proprioception might well be the only signs in a paramedian brainstem lesion and in this case you cannot definitively localise the lesion at the bedside.

If you find abnormalities of pain and temperature sensation in the arm and leg on the same side as the weakness, then the problem is almost certainly not in the brainstem unless there is widespread brainstem ischaemia involving both the paramedian and long circumferential branches—for example, basilar

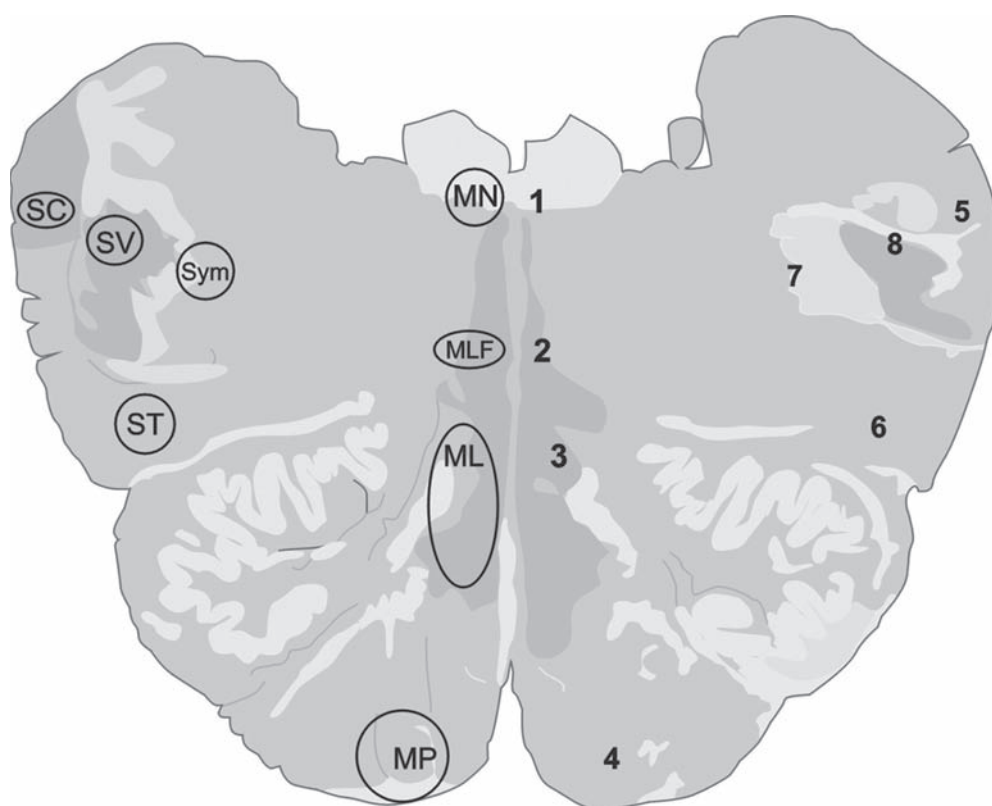


Figure 2

A cross section of the brainstem (at the level of the medulla) showing the four structures in the medial and the four structures in the lateral part of the brainstem. The size of the outlined areas is not representative of their anatomical size. (1) MN, motor nucleus (3, 4, 6 or 12); (2) MLF, median longitudinal fasciculus; (3) ML, medial lemniscus; (4) MP, motor pathway (corticospinal/pyramidal tract); (5) SC, spinocerebellar; (6) St, spinothalamic; (7) Sym, sympathetic; (8) Sv, sensory nucleus of the fifth cranial nerve. (reproduced from Gates⁴).

artery thrombosis—because the pathway for pain and temperature is in the lateral, rather than the paramedian brainstem (see below).

You would not expect to detect any cerebellar signs because the cerebellar pathways are in the lateral aspect of the brainstem but remember that weak limbs can look 'ataxic', due to the weakness. It is any associated cranial nerve abnormalities that would clearly indicate the problem is in the brainstem. There are three cranial nerves—the third in the midbrain, sixth in the pons and 12th in the medulla—that emerge just lateral to the motor pathway and therefore may be affected by, for example, ischaemia in the distribution of the small paramedian perforating vessels. Therefore, if the patient you are examining with the hemiparesis has a third, sixth (paramedian pontine lesions are rare) or 12th cranial nerve palsy contralateral to the side of weakness, the problem must be in the paramedian midbrain, pons or medulla, respectively, on the side of the cranial nerve palsy (figure 3).

Although the medial longitudinal fasciculus is in the paramedian brainstem, it is further posterior in the brainstem and is usually not affected by occlusion of the paramedian perforating vessels. Isolated involvement of the medial longitudinal fasciculus in multiple sclerosis and lacunar infarcts cause an 'internuclear' ophthalmoplegia (figure 4).

ANATOMY OF THE LATERAL BRAINSTEM

All four structures in the lateral aspect (or the side) of the brainstem begin with the letter S (figures 2):

- Spinothalamic pathway
- Sympathetic pathway
- Sensory nucleus of the fifth cranial nerve
- Spinocerebellar tract

The spinothalamic pathway for pain and temperature crosses the midline immediately it enters the spinal cord, travels caudally to the lateral aspect of the brainstem and then on to the thalamus.

The sympathetic pathway extends rostrally from the hypothalamus via the lateral brainstem into the ipsilateral spinal cord.

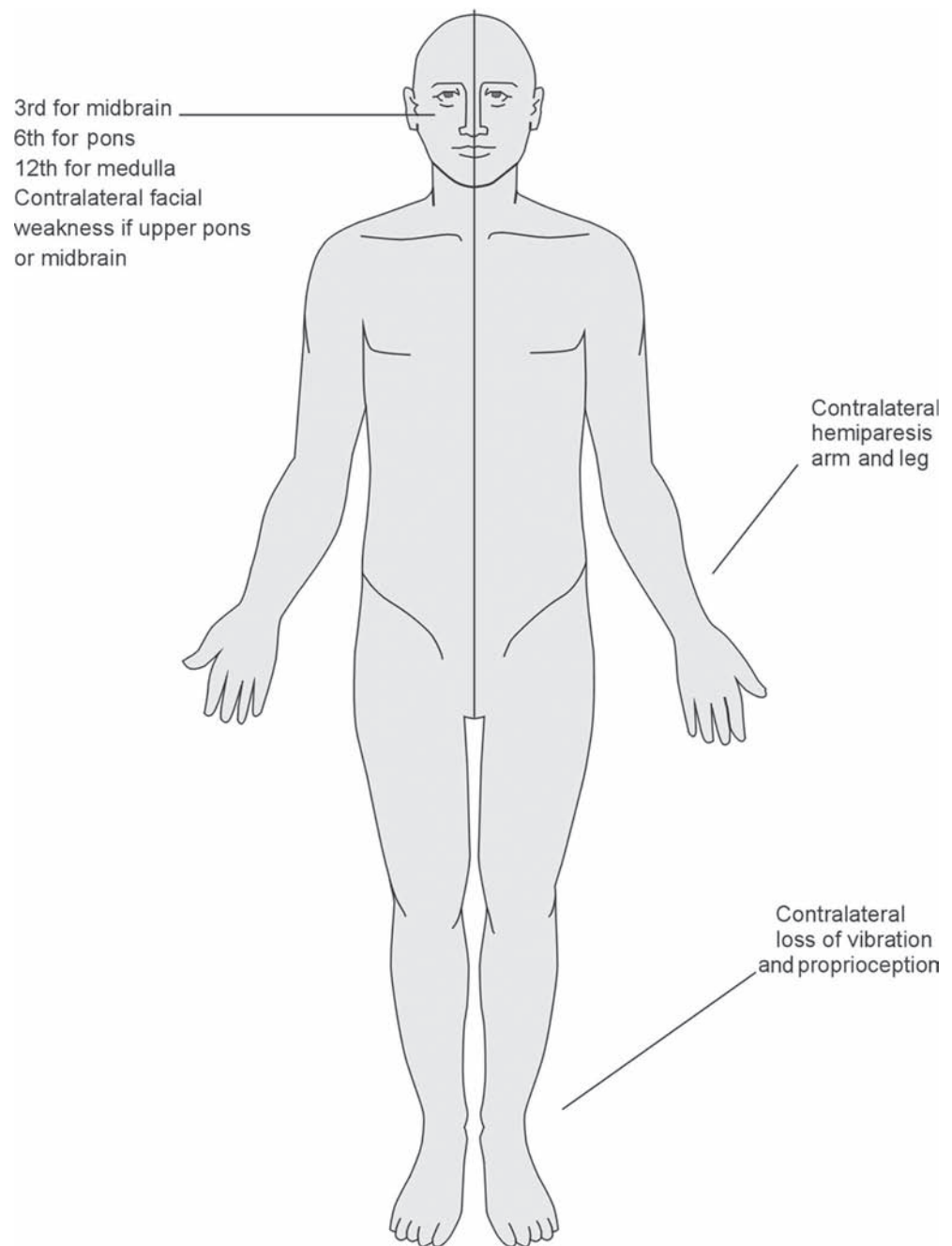
The sensory (trigeminal) nucleus of the fifth cranial nerve nucleus extends throughout the lateral brainstem from the midbrain to the medulla, and continues into the cervical cord where it merges with the dorsal horn cells.

The spinocerebellar tract is a set of fibres originating in the spinal cord and terminating in the ipsilateral cerebellum; this pathway also lies in the lateral brainstem.

Note that all four structures extend throughout the length of the brainstem.

Figure 3

The signs of a right paramedian brainstem syndrome (reproduced from Gates⁴).

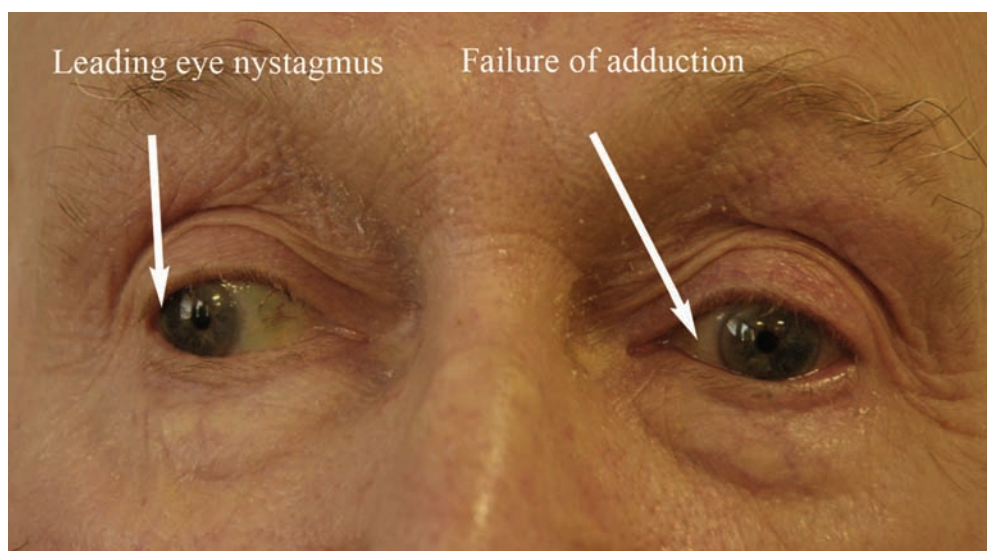


PATIENT NO 2

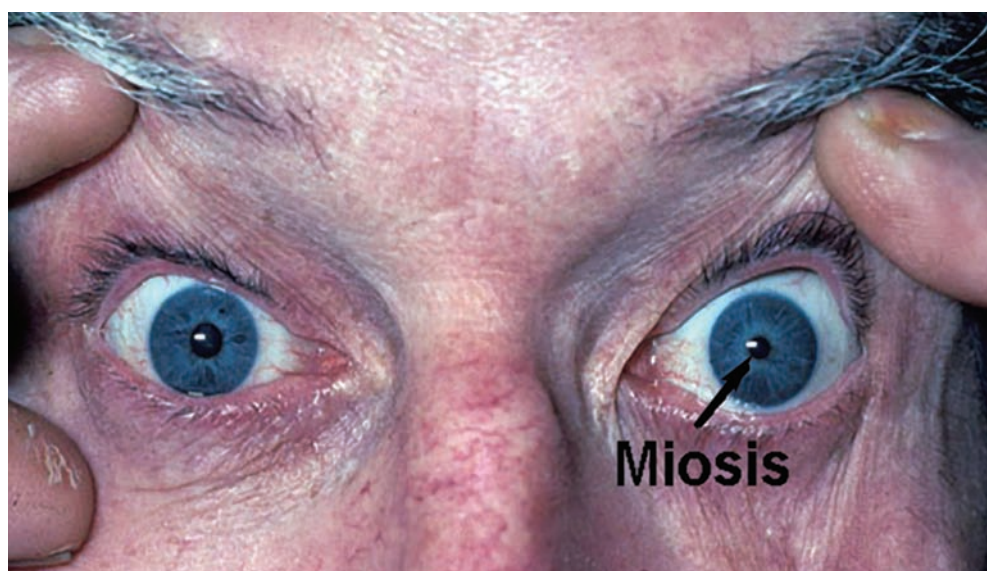
Let us now assume that you see another patient; the examination of the motor system does not reveal any abnormality and therefore the problem cannot be in the paramedian brainstem. If it is not a paramedian brainstem syndrome then you would not anticipate any abnormality of vibration and proprioception when you turn to the sensory examination. But if your examination detects impaired pain and temperature sensation in one arm and leg, then you know that the problem is in the spinothalamic pathway on the opposite side to the impaired sensation, and if the problem

is in the brainstem it has to be in the lateral aspect. If there is impaired pain and temperature sensation affecting the face on the same side as the arm and leg, then the problem is likely to be above the fifth nerve nucleus and therefore likely to be above the level of the brainstem.

Continuing your examination of the arms and legs you detect ataxia affecting the arm and leg on the opposite side to the impaired pain and temperature sensation in the limbs. This is due to involvement of the ipsilateral (to the ataxia) spinocerebellar pathway and together these signs clearly indicate a brainstem problem.

**Figure 4**

A left internuclear ophthalmoplegia. Gaze to the right: failure of adduction of the left eye towards the nose with leading (abducting) eye nystagmus in the right eye (reproduced from Gates⁴).

**Figure 5**

A left Horner's syndrome. The ptosed left eyelid is being elevated to reveal the miosis. (reproduced from Gates⁴).

Turning now to the cranial nerves, if you find impaired pain and temperature sensation on the face contralateral to the pain and temperature loss in the arm and leg, this is due to involvement of the sensory nucleus of the fifth cranial nerve. The distribution of the sensory loss will be within the fifth cranial nerve territory, affecting the face to the midline, the forehead and scalp to the junction of the anterior two-thirds and posterior one-third, the temple and the anterior aspect of the earlobe, the cheek and jaw but it will spare the angle of the jaw which is supplied by the second and third cervical sensory roots.

The presence of impaired pain and temperature on one side of the face and the opposite side of the body is referred to as 'crossed' sensory signs and is only seen in lateral brainstem lesions.

You may find ptosis and miosis (Horner's syndrome, figure 5) on the same side as the facial sensory loss due to involvement of the sympathetic pathway. At this stage you have detected the signs of a lateral brainstem syndrome but you cannot localise it to the pons or the medulla (lateral brainstem syndromes do not occur in the midbrain) (figure 6).

In some patients with lateral brainstem syndromes, this is all you find and thus it is impossible clinically to localise the problem to a specific level in the brainstem. However, the presence of hoarseness, dysarthria, dysphagia, impaired sensation of the throat and weakness of palatal elevation on the side of the facial sensory loss clearly indicates involvement of the ninth, 10th and 11th cranial nerves in the lateral medulla, while a lower motor neuron facial weakness and weakness of the masseter

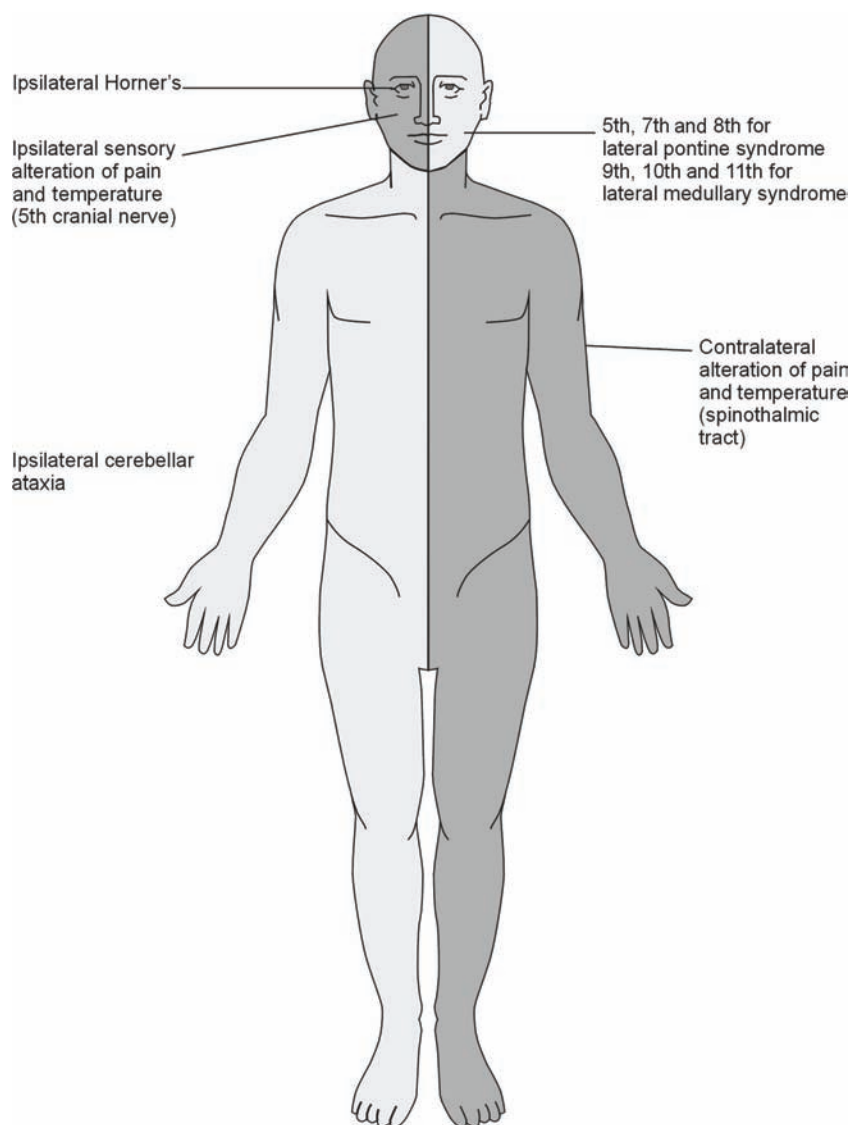


Figure 6
The signs seen in a right lateral brainstem lesion (reproduced from Gates⁴).

and pterygoid muscles on the same side as the facial sensory loss points to involvement of the fifth motor and seventh cranial nerves in the lateral pons. Although the eighth cranial nerve is in the lateral pons, it is rarely affected. Vertigo and complex disturbances of ocular movements may occur with lateral

medullary lesions² but are beyond the scope of this article.

WHICH ALL LEADS TO 'THE RULE OF 4'

- There are *four* structures in the midline starting with the letter M.
- There are *four* structures in the lateral brainstem starting with the letter S.
- The lower *four* cranial nerves are in the medulla, the middle *four* cranial nerves are in the pons and the first *four* cranial nerves are above the pons, with the third and fourth in the midbrain.
- The *four* motor cranial nerves that are in the midline are the four that divide evenly into 12 (except for 1 and 2)—that is, 3, 4, 6 and 12.

I have also explained this simple technique for remembering the neuroanatomy of the brainstem in a prior publication.³

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